

A two-branched numerical solution of the two-dimensional Bratu's problem

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ABSTRACT

We intend to numerically solve the famous two-dimensional Bratu's problem through developing a new iterative finite difference algorithm. The proposed scheme is capable of accurately approximating both branches of the solution of the considered problem. We first introduce a new transformation of Bratu's problem conserving the solution bifurcated behavior. Then, using Newton-Raphson-Kantorovich approximation in function space, we develop an iterative finite difference method yielding a simple algorithm for approximating the sequence of numerical solutions. We also perform a convergence analysis proving that our algorithm converges quadratically to the exact solution of the problem. Finally, we present numerical simulation results showing the capability of our algorithm to accurately compute the two-branches of the solution for the two-dimensional Bratu's problem.

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1. Introduction

Bratu's problem, also known as Liouville-Bratu-Gelfand equation [11,20,26], is a strongly nonlinear Poisson's equation that arises in a variety of physical and chemical phenomena, such as chemical reactor theory, radiative heat transfer, the Chandrasekhar model of the universe expansion, and in various nano-technology applications [2,15]. It also arises in a wide range of engineering applications such as the solid fuel ignition model describing the thermal reaction process in combustible non-deformable materials [22], in addition to the electro-spinning process for the production of ultra-fine polymer fibers [37]. Furthermore, due to its simplicity, Bratu's problem is used to test various nonlinear numerical solvers [6]. In this manuscript, we investigate the following two-dimensional Bratu's problem

$$\Delta u + \lambda e^u = 0, \quad \text{for } (x, y) \in \Omega, \quad (1a)$$

subject to the homogeneous Dirichlet boundary conditions

$$u(x, y) = 0, \quad \text{for } (x, y) \in \Gamma = \partial\Omega, \quad (1b)$$

where Ω is a two-dimensional rectangular domain in $\mathbb{R}^2$, $\Gamma = \partial\Omega$ is the boundary of Ω , and $\lambda > 0$ is a positive real number called the transition parameter of Bratu's problem.

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It is noteworthy to mention that since all the singular points are limit points, Bratu's problem has no bifurcation points [15]. For more details about the applications of Bratu's problem and the solution behavior, consult [8,20,21,23,29,35]. Furthermore, it is known according to the aforementioned studies that Bratu's problem has no solution when λ is greater than a certain value λ_c called the *critical value* of the transition parameter. Nevertheless, it has two solutions for $\lambda < \lambda_c$ and only one solution for $\lambda = \lambda_c$.

In the one dimensional case, various analytical methods and numerical techniques have been utilized to solve Bratu's equation [1,3,4,8,9,12–14,17,24,25,28,29,34–36]. It has been numerically shown that the critical value of the transition parameter in that case is $\lambda_c = 3.513830719$. However, only a few studies have investigated the two-dimensional case of Bratu's problem. For instance, we can cite the work of Boyd [10] who performed both analytical and numerical studies of the two-dimensional Bratu's problem. Also, Chang and Chien [16] studied the problem numerically by generally solving nonlinear elliptic eigenvalue problems containing folds and bifurcation points. In this latter work, the authors proposed a robust algorithm based on a multigrid-Lanczos method which is able to track solution branches of discrete problems and to detect singular points along the branches of the solution. In addition, Mohsen et al. [30] studied Bratu's problem by enhancing multigrid-based methods using a new smoother; whereas, Doedel [18] tackled the problem by developing collocation methods for continuation problems. Also, Fedoseyev et al. [19] treated the problem through a meshless collocation method known as the multi-quadric radial basis function method. Some other studies such as [27,38] and [31] used various methods like the weighted residual method, the finite difference method, Coiflet-type wavelet expansion, and some analytical methods to solve the two-dimensional Bratu's problem. On the other hand, with regard to the critical value of the transition parameter λ , different numerical studies have lead to different values. For instance, Liu et al. [27] found out that the critical value λ_c lies between 6.80 and 6.81. Other studies have found more accurate value of the critical value of the transition parameter as $\lambda_c = 6.808124423$ [16,18,19,30]. However, Odejide et al. [31] predict the critical value of the transition parameter to be between 6.939571823 and 7.12222.

In the present manuscript, we develop an iterative finite difference method that can compute accurate numerical solution of Bratu's problem in the two-dimensional case, for any value of the transition parameter such that $\lambda \leq \lambda_c$. We also numerically estimate the critical value of the transition parameter λ_c . This work can be considered as an extension of our recent work on the one-dimensional Bratu's problem [8,35] in which we used a similar approach to approximate both branches of the solutions in that case. For this purpose, we first introduce a transformation of Bratu's problem into a new simpler nonlinear boundary-value problem. Then, we perform a quasilinearization of the transformed problem using Newton-Raphson-Kantorovich approximation in function space [7] leading to a sequence of approximate solutions. Such a sequence is shown to be quadratically convergent. Furthermore, in order to numerically solve for the sequence of approximate solutions, we use a finite difference method on the quasilinearized equation. Based on these steps, we numerically demonstrate that our method is successful in computing the two branches of the solution of the two-dimensional Bratu's problem. To the best of our knowledge, very few of the published numerical schemes were able to efficiently solve Bratu's problem and fewer were able to accurately estimate the value of the critical value λ_c . We believe that ours is an effective and promising method for computing the critical value of the transition parameter and for approximating the two branches of the solution of Bratu's problem.

This manuscript is structured as follows. In Section 2, we introduce a new transformation function allowing us to rewrite Bratu's problem in a more convenient form. An iterative finite difference method is then developed to numerically approximate the transformed nonlinear partial differential equation. In Section 3, we investigate how to choose the initial guess in order to guarantee that the scheme converges to the desired branch of the solution. We also perform a thorough convergence analysis in Section 4. Finally, we illustrate several numerical experiments showing the accuracy of our method in Section 5.

2. Iterative finite difference method for the two-dimensional problem

2.1. A transformation of the two-dimensional Bratu's problem

First, let us introduce a new transformation function which will allow us to rewrite Bratu's problem (1) as a simpler nonlinear ordinary differential equation, while maintaining the bifurcated behavior of the original solution. Hence, we define the following alternative function

$$v = e^{-\frac{1}{2}u}. \quad (2)$$

Equation (2) gives

$$u(x, y) = -2 \ln(v(x, y)), \quad (3)$$

which has the following partial derivatives

$$u_x = -2 \frac{v_x}{v}, \quad u_y = -2 \frac{v_y}{v}, \quad u_{xx} = -2 \frac{v_{xx}v - v_x^2}{v^2}, \quad u_{yy} = -2 \frac{v_{yy}v - v_y^2}{v^2}, \quad (4)$$

where v_x and v_y denote the first order partial derivatives of v with respect to x and y , respectively; while v_{xx} and v_{yy} denote the second order partial derivatives of v with respect to x and with respect to y , respectively.

Plugging (4) into (1) and using the fact that $v^2 = e^{-u}$ (i.e. $e^u = v^{-2}$), we obtain

$$-2 \frac{v_{xx}v - v_x^2 + v_{yy}v - v_y^2}{v^2} + \lambda v^{-2} = 0, \quad (5)$$

which can be simplified as

$$-2 \left(v_{xx}v - v_x^2 + v_{yy}v - v_y^2 \right) + \lambda = 0. \quad (6)$$

As a result, the following transformed two-dimensional Bratu's boundary-value problem is obtained:

$$v_{xx}v - v_x^2 + v_{yy}v - v_y^2 - \frac{\lambda}{2} = 0, \quad \text{over } (0, 1) \times (0, 1), \quad (7a)$$

subject to the following boundary conditions

$$v(0, y) = 1, \quad v(1, y) = 1, \quad v(x, 0) = 1, \quad \text{and} \quad v(x, 1) = 1. \quad (7b)$$

2.2. Iterative finite difference method for the transformed problem

After transforming Bratu's problem (1) into its simplified form (7), we use the Newton-Raphson-Kantorovich quasilinearization technique in order to establish an efficient numerical solver of the transformed problem (7). This technique is an extension into the two-dimensional case of the iterative finite difference scheme applied on the one-dimensional Bratu's problem in [8].

First, we rewrite Equation (7a) as $f(v, v_x, v_y, v_{xx}, v_{yy}) = 0$, where

$$f(v, v_x, v_y, v_{xx}, v_{yy}) = v_{xx}v - v_x^2 + v_{yy}v - v_y^2 - \frac{\lambda}{2}. \quad (8)$$

Next, we construct a new sequence $\{v^n\}$ of approximate solutions of the nonlinear partial differential equation $f(v, v_x, v_y, v_{xx}, v_{yy}) = 0$. To do this, we extend the Newton-Raphson-Kantorovich approximation method in function space (originally formulated by Bellman and Kalaba [7] for ordinary differential equations) to the case of partial differential equations of the form $f(v, v_x, v_y, v_{xx}, v_{yy}) = 0$, where f is differentiable with respect to v, v_x, v_y, v_{xx} , and v_{yy} . The resulting generalized Newton-Raphson-Kantorovich approximation can be elaborated as

$$\begin{aligned} f(v^{n+1}, v_x^{n+1}, v_y^{n+1}, v_{xx}^{n+1}, v_{yy}^{n+1}) &\approx f(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v^{n+1} - v^n) f_v(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_x^{n+1} - v_x^n) f_{v_x}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_y^{n+1} - v_y^n) f_{v_y}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_{xx}^{n+1} - v_{xx}^n) f_{v_{xx}}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_{yy}^{n+1} - v_{yy}^n) f_{v_{yy}}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n), \end{aligned}$$

which yields the iterative scheme below

$$\begin{aligned} f(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) &\quad + (v^{n+1} - v^n) f_v(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_x^{n+1} - v_x^n) f_{v_x}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_y^{n+1} - v_y^n) f_{v_y}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_{xx}^{n+1} - v_{xx}^n) f_{v_{xx}}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) \\ &\quad + (v_{yy}^{n+1} - v_{yy}^n) f_{v_{yy}}(v^n, v_x^n, v_y^n, v_{xx}^n, v_{yy}^n) = 0. \end{aligned} \quad (9)$$

In Section 4 we will prove that the sequence $\{v^n\}$ solution of the scheme (9) converges to the solution of the nonlinear PDE $f(v, v_x, v_y, v_{xx}, v_{yy}) = 0$.

By computing the derivatives of the functional $f(v, v_x, v_y, v_{xx}, v_{yy}) = 0$ defined in (8), we obtain:

$$\begin{aligned} f_v(v, v_x, v_y, v_{xx}, v_{yy}) &= v_{xx} + v_{yy}, \\ f_{v_x}(v, v_x, v_y, v_{xx}, v_{yy}) &= -2v_x, \\ f_{v_y}(v, v_x, v_y, v_{xx}, v_{yy}) &= -2v_y, \\ f_{v_{xx}}(v, v_x, v_y, v_{xx}, v_{yy}) &= v, \\ f_{v_{yy}}(v, v_x, v_y, v_{xx}, v_{yy}) &= v. \end{aligned} \quad (10)$$

Plugging the derivatives (10) into (9) yields the following scheme

$$v_{xx}^n v^n - (v_x^n)^2 + v_{yy}^n v^n - (v_y^n)^2 - \frac{\lambda}{2} + (v^{n+1} - v^n)(v_{xx}^n + v_{yy}^n) - 2v_x^n(v_x^{n+1} - v_x^n) - 2v_y^n(v_y^{n+1} - v_y^n) + v^n(v_{xx}^{n+1} - v_{xx}^n) + v^n(v_{yy}^{n+1} - v_{yy}^n) = 0, \quad (11)$$

which is then simplified as follows

$$(v_{xx}^n + v_{yy}^n) v^{n+1} - 2v_x^n v_x^{n+1} - 2v_y^n v_y^{n+1} + v^n v_{xx}^{n+1} + v^n v_{yy}^{n+1} = \frac{\lambda}{2} + v^n (v_{xx}^n + v_{yy}^n) - (v_x^n)^2 - (v_y^n)^2, \quad (12a)$$

with boundary conditions

$$v^{n+1}(0, y) = 1, \quad v^{n+1}(1, y) = 1, \quad v^{n+1}(x, 0) = 1, \quad \text{and} \quad v^{n+1}(x, 1) = 1. \quad (12b)$$

Hence, based on the established scheme (12), the function v^{n+1} can be computed as the solution of the linear PDE (12a) given the function v^n and the boundary conditions (12b). In our case, we solve the linear PDE (12a) at every iteration, using the standard finite difference method. Furthermore, in this regard, it is important to properly choose the initial guess v^0 in order to guarantee the convergence of the scheme (12) to the correct solution of the problem. In Section 3, we discuss how to choose a convenient initial guess function for the proposed scheme.

Hence, let $\{(x_i, y_j)\}_{i=0 \dots M_1, j=0 \dots M_2}$ be a mesh of size $M_1 \times M_2$ over the square domain $[0, 1] \times [0, 1]$, with step sizes $h_x = 1/M_1$ and $h_y = 1/M_2$ in the x and y directions, respectively. For a given n , let V^n be the finite difference solution. Without loss of generality, we assume that $h_x = h_y = h$ and $M_1 = M_2 = M$.

For the numerical approximation of the first order partial derivatives (with respect to x or to y), we use the known standard centered finite difference approximations of the first derivative as follows

$$v_x^{n+1}(x_i, y_j) \approx D_x V^{n+1}(x_i, y_j) = \frac{V^{n+1}(x_{i+1}, y_j) - V^{n+1}(x_{i-1}, y_j)}{2h}, \quad (13a)$$

$$v_y^{n+1}(x_i, y_j) \approx D_y V^{n+1}(x_i, y_j) = \frac{V^{n+1}(x_i, y_{j+1}) - V^{n+1}(x_i, y_{j-1})}{2h}, \quad (13b)$$

while for the approximation of the second order partial derivatives, we use the following standard centered finite difference approximations of the second derivative

$$v_{xx}^{n+1}(x_i, y_j) \approx D_{xx} V^{n+1}(x_i, y_j) = \frac{V^{n+1}(x_{i+1}, y_j) - 2V^{n+1}(x_i, y_j) + V^{n+1}(x_{i-1}, y_j)}{h^2}, \quad (13c)$$

$$v_{yy}^{n+1}(x_i, y_j) \approx D_{yy} V^{n+1}(x_i, y_j) = \frac{V^{n+1}(x_i, y_{j+1}) - 2V^{n+1}(x_i, y_j) + V^{n+1}(x_i, y_{j-1})}{h^2}, \quad (13d)$$

all for $1 \leq i \leq M-1$ and $1 \leq j \leq M-1$.

For simplicity, we will use in the remaining part of this section the following abbreviated notations $V^n(x_i, y_j) = V_{i,j}^n$, $D_x V^{n+1}(x_i, y_j) = D_x V_{i,j}^{n+1}$, $D_y V^{n+1}(x_i, y_j) = D_y V_{i,j}^{n+1}$, $D_{xx} V^{n+1}(x_i, y_j) = D_{xx} V_{i,j}^{n+1}$, and $D_{yy} V^{n+1}(x_i, y_j) = D_{yy} V_{i,j}^{n+1}$.

After substituting Equations (13) into (12a) and performing the required simplifications, we obtain

$$\begin{aligned} & \left(\frac{V_{i,j}^n}{h^2} - \frac{D_x V_{i,j}^n}{h} \right) V_{i+1,j}^{n+1} + \left(\frac{V_{i,j}^n}{h^2} - \frac{D_y V_{i,j}^n}{h} \right) V_{i,j+1}^{n+1} \\ & + \left(D_{xx} + D_{yy} - 4 \frac{V_{i,j}^n}{h^2} \right) V_{i,j}^{n+1} \\ & + \left(\frac{V_{i,j}^n}{h^2} + \frac{D_x V_{i,j}^n}{h} \right) V_{i-1,j}^{n+1} + \left(\frac{V_{i,j}^n}{h^2} + \frac{D_y V_{i,j}^n}{h} \right) V_{i,j-1}^{n+1} \\ & = \frac{\lambda}{2} + V_{i,j}^n (D_{xx} V_{i,j}^n + D_{yy} V_{i,j}^n) - (D_x V_{i,j}^n)^2 - (D_y V_{i,j}^n)^2, \end{aligned} \quad (14a)$$

for $1 \leq i \leq M-1$ and $1 \leq j \leq M-1$.

Furthermore, the boundary conditions are simply discretized as

$$V^{n+1}(0, y_j) = 1, \quad V^{n+1}(1, y_j) = 1, \quad V^{n+1}(x_i, 0) = 1, \quad \text{and} \quad V^{n+1}(x_i, 1) = 1, \quad (14b)$$

for $1 \leq i \leq M-1$ and $1 \leq j \leq M-1$.

3. Choosing the initial guess function v^0

In this section, we investigate the crucial choice of the initial guess function v_0 guaranteeing the convergence of the proposed scheme to the correct solution(s) of the problem. Here, we choose an initial approximation satisfying the boundary conditions (7b) at the corners of the domain, which has the following form:

$$v^0(x, y) = 2 - (1 + x - x^2)^a(1 + y - y^2)^a, \quad a > 0, \quad (15)$$

where a is a positive parameter to be determined as follows.

First, we compute the partial derivatives of v^0

$$\begin{aligned} v_x^0(x, y) &= -a(1 - 2x)(1 + x - x^2)^{a-1}(1 + y - y^2)^a, \\ v_y^0(x, y) &= -a(1 - 2y)(1 + x - x^2)^a(1 + y - y^2)^{a-1}, \\ v_{xx}^0(x, y) &= -a(1 + y - y^2)^a \left[-2(1 + x - x^2)^{a-1} + (a - 1)(1 - 2x)^2(1 + x - x^2)^{a-2} \right], \\ v_{yy}^0(x, y) &= -a(1 + x - x^2)^a \left[-2(1 + y - y^2)^{a-1} + (a - 1)(1 - 2y)^2(1 + y - y^2)^{a-2} \right]. \end{aligned}$$

Let us call $r = 1 + x - x^2$ and $s = 1 + y - y^2$ and using the following equality

$$(1 - 2x)^2 = 5 - 4(1 + x - x^2),$$

gives

$$v_{xx}^0(x, y) = (4a^2 - 2a)r^{a-1}s^a - 5(a^2 - a)r^{a-2}s^a, \quad (16)$$

$$v_{yy}^0(x, y) = (4a^2 - 2a)r^as^{a-1} - 5(a^2 - a)r^as^{a-2}, \quad (17)$$

and

$$v_x^0(x, y)^2 = 5a^2r^{2a-2}s^{2a} - 4a^2r^{2a-1}s^{2a}, \quad (18)$$

$$v_y^0(x, y)^2 = 5a^2r^{2a}s^{2a-2} - 4a^2r^{2a}s^{2a-1}. \quad (19)$$

Next, we substitute the partial derivatives v_x^0 , v_y^0 , v_{xx}^0 , and v_{yy}^0 into the transformed problem (7) which leads (after simplification) to

$$\begin{aligned} &-(10a^2 - 10a)(rs)^{a-2}(s^2 + r^2) - 5a(rs)^{2a-2}(s^2 + r^2) \\ &+ (8a^2 - 4a)(rs)^{a-1}(s + r) + 2a(rs)^{2a-1}(s + r) = \frac{\lambda}{2}. \end{aligned} \quad (20)$$

We note that

$$\min_{0 \leq x \leq 1} r(x) = r(0) = r(1) = 1 \quad \text{and} \quad \max_{0 \leq x \leq 1} r(x) = r\left(\frac{1}{2}\right) = \frac{5}{4},$$

and similarly, for $s(y)$.

Therefore, given the symmetry of the problem and for sake of simplicity, without loss of generality, we introduce a variable t replacing both r and s for $0 \leq x, y \leq 1$. Hence, we have $t \in [1, 1.25]$ and Equation (20) becomes

$$\begin{aligned} &-(10a^2 - 10a)t^{2(a-2)}(2t^2) - 5at^{2(2a-2)}(2t^2) \\ &+ (8a^2 - 4a)t^{2(a-1)}(2t) + 2at^{2(2a-1)}(2t) = \frac{\lambda}{2}, \end{aligned} \quad (21)$$

which can also be simplified as

$$a(4t - 10)t^{4a-2} + (16a^2 - 8a)t^{2a-1} - 20(a^2 - a)t^{2a-2} = \frac{\lambda}{2}. \quad (22)$$

We now let

$$W(a, t, \lambda) = a(4t - 10)t^{4a-2} + (16a^2 - 8a)t^{2a-1} - 20(a^2 - a)t^{2a-2} - \frac{\lambda}{2}.$$

Then, in order to compute the values the parameter a guaranteeing the convergence of the iterative scheme to upper and lower solutions of the transformed Bratu's problem, we plot the zeros level curves of $W(a, t, \lambda)$ for different values of λ in Fig. 1.

Fig. 1 shows that a choice of $a = 0.25$ will guarantee the convergence of the iterative scheme to the lower solution and a choice of $a = 1.25$ will guarantee the convergence to the upper solution regardless of the value of λ . Additionally, a choice of $a = 0.75$ will guarantee the convergence to the unique solution of Bratu's problem when $\lambda = \lambda_c$.

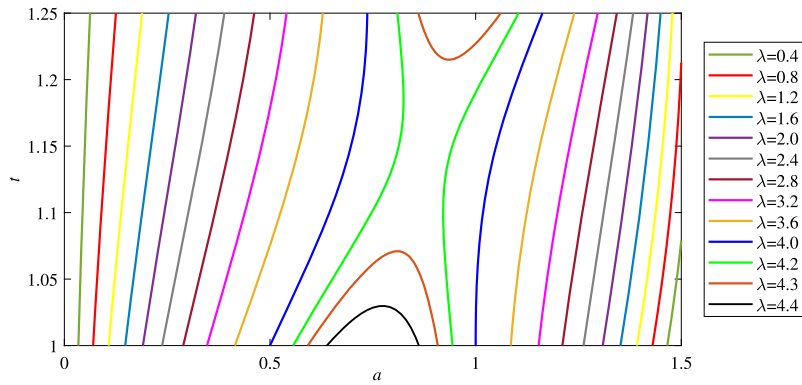


Fig. 1. The convergence condition on the initial solutions for different values of λ . (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

4. Convergence analysis

Here, we first show that the sequence $\{v^n\}$ of approximate solutions of (7) is well defined. We also prove that it converges quadratically to the exact solution v . For that purpose and for simplicity, let us consider the following nonlinear second-order generalized PDE

$$u_{xx} + u_{yy} = g(u, u_x, u_y), \quad (x, y) \in (0, b)^2, \quad (23a)$$

subject to the boundary conditions

$$u(0, y) = u(b, y) = u(x, 0) = u(x, b) = 1. \quad (23b)$$

We introduce an initial guess $u^0(x, y)$ and we consider also a sequence of functions $\{u^n(x, y)\}$ given by the following relation of recurrence

$$\begin{aligned} u_{xx}^{n+1} + u_{yy}^{n+1} = & g(u^n, u_x^n, u_y^n) + (u^{n+1} - u^n)g_u(u^n, u_x^n, u_y^n) + (u_x^{n+1} - u_x^n)g_{u_x}(u^n, u_x^n, u_y^n) \\ & + (u_y^{n+1} - u_y^n)g_{u_y}(u^n, u_x^n, u_y^n), \quad (x, y) \in (0, b)^2, \end{aligned} \quad (24a)$$

subject to the boundary conditions

$$u_{n+1}(0, y) = u_{n+1}(b, y) = u_{n+1}(x, 0) = u_{n+1}(x, b) = 1. \quad (24b)$$

Since, the right-hand side of Equation (24a) is linear with respect to u^{n+1} , u_x^{n+1} , and u_y^{n+1} , then a very important advantage of our algorithm is that every function $u^{n+1}(x, y)$ is the solution of a linear partial differential equation. Now, we demonstrate that the sequence $\{u^n(x, y)\}$ is well defined and that it uniformly converges to u . We also prove that the convergence is quadratic using the maximum norm.

Next, we establish a theorem stating that, for sufficiently small b , the sequence $\{u^n\}$ is well defined. Hence, let us use the notation $C_b^m(D)$ to designate the set of real m -times continuously differentiable functions which are bounded together with their derivatives up to the order m .

Theorem 4.1. Suppose that the function $g(u, u_x, u_y) \in C_b^1(\mathbb{R})$. Then the sequence $\{u_n\}$ defined by (24) is well defined provided that b is sufficiently small.

Proof. Similarly to the proof in [7], we demonstrate the existence and the uniform boundedness of the sequence $\{u^n\}$ for b sufficiently small. We let $K(x, y; \xi, \eta)$ to be the Green's function associated with the BVP: $-v_{xx} - v_{yy} = \phi(x, y)$, subject to: $v(0, y) = v(b, y) = v(x, 0) = v(x, b) = 0$, such that

$$v(x, y) = \int_0^b \int_0^b K(x, y; \xi, \eta) \phi(\xi, \eta) d\xi d\eta.$$

The function $K(x, y; \xi, \eta)$ is given explicitly by

$$K(x, y; \xi, \eta) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \frac{1}{n^2 + m^2} \sin\left(\frac{n\pi x}{b}\right) \sin\left(\frac{n\pi \xi}{b}\right) \sin\left(\frac{m\pi y}{b}\right) \sin\left(\frac{m\pi \eta}{b}\right). \quad (25)$$

Due to its linearity, the solution to the BVP: $-u_{xx} - u_{yy} = \phi(x, y)$, subject to: $u(0, y) = u(b, y) = u(x, 0) = u(x, b) = 1$ is

$$u(x, y) = 1 + \int_0^b \int_0^b K(x, y; \xi, \eta) \phi(\xi, \eta) d\xi d\eta.$$

We note that (24) can be viewed as a BVP of the form $u_{xx} + u_{yy} = -\phi(x, y)$ subject to $u(0, y) = u(b, y) = u(x, 0) = u(x, b) = 1$. Thus, we can write

$$\begin{aligned} u^{n+1} = 1 + \int_0^b \int_0^b K(x, y; \xi, \eta) & [g(u^n, u_x^n, u_y^n) + (u^{n+1} - u^n)g_u(u^n, u_x^n, u_y^n) \\ & + (u_x^{n+1} - u_x^n)g_{u_x}(u^n, u_x^n, u_y^n) \\ & + (u_y^{n+1} - u_y^n)g_{u_y}(u^n, u_x^n, u_y^n)] d\xi d\eta. \end{aligned} \quad (26)$$

Let

$$K_1 = \max_{0 \leq x, y, \xi, \eta \leq b} |K(x, y; \xi, \eta)| < \infty, \quad (27a)$$

$$K_2 = \max_{0 \leq x, y, \xi, \eta \leq b} |K_x(x, y; \xi, \eta)| < \infty, \quad (27b)$$

$$K_3 = \max_{0 \leq x, y, \xi, \eta \leq b} |K_y(x, y; \xi, \eta)| < \infty. \quad (27c)$$

We choose the initial guess $u^0(x, y)$ such that $|u^0(x, y)| \leq 2$, $|u_x^0(x, y)| \leq 1$, and $|u_y^0(x, y)| \leq 1$ for all $(x, y) \in [0, b]^2$. Then, we let

$$m = \max \left(\max_D |g(u, u_x, u_y)|, \max_D |g_u(u, u_x, u_y)|, \max_D |g_{u_x}(u, u_x, u_y)|, \max_D |g_{u_y}(u, u_x, u_y)| \right),$$

where $D = \{(u, u_x, u_y) \mid |u| \leq 2, |u_x| \leq 1, |u_y| \leq 1\}$.

If $g(u, u_x, u_y) \in C_b^2(\mathbb{R})$ then $m < \infty$. From (26), we have

$$\begin{aligned} |u^{n+1}| & \leq 1 + \int_0^b \int_0^b |K(x, y; \xi, \eta)| [|g(u^n, u_x^n, u_y^n)| + |u^{n+1} - u^n| |g_u(u^n, u_x^n, u_y^n)| \\ & + |u_x^{n+1} - u_x^n| |g_{u_x}(u^n, u_x^n, u_y^n)| + |u_y^{n+1} - u_y^n| |g_{u_y}(u^n, u_x^n, u_y^n)|] d\xi d\eta \\ & \leq 1 + mK_1 \int_0^b \int_0^b [1 + |u^{n+1}| + |u^n| + |u_x^{n+1}| + |u_x^n| + |u_y^{n+1}| + |u_y^n|] d\xi d\eta \\ & \leq 1 + mK_1 b^2 \left[1 + \max_{x,y} |u^{n+1}| + \max_{x,y} |u^n| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_x^n| \right. \\ & \quad \left. + \max_{x,y} |u_y^{n+1}| + \max_{x,y} |u_y^n| \right]. \end{aligned}$$

Taking the maximum on both sides, we get

$$\begin{aligned} \max_{x,y} |u^{n+1}| & \leq 1 + mK_1 b^2 \left[1 + \max_{x,y} |u^{n+1}| + \max_{x,y} |u^n| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_x^n| \right. \\ & \quad \left. + \max_{x,y} |u_y^{n+1}| + \max_{x,y} |u_y^n| \right]. \end{aligned} \quad (28)$$

On the other hand, we have from (26)

$$\begin{aligned} u_x^{n+1} = \int_0^b \int_0^b K_x(x, y; \xi, \eta) & [g(u^n, u_x^n, u_y^n) + (u^{n+1} - u^n)g_u(u^n, u_x^n, u_y^n) \\ & + (u_x^{n+1} - u_x^n)g_{u_x}(u^n, u_x^n, u_y^n) + (u_y^{n+1} - u_y^n)g_{u_y}(u^n, u_x^n, u_y^n)] d\xi d\eta, \end{aligned}$$

and

$$u_y^{n+1} = \int_0^b \int_0^b K_y(x, y; \xi, \eta) \left[g(u^n, u_x^n, u_y^n) + (u^{n+1} - u^n)g_u(u^n, u_x^n, u_y^n) \right. \\ \left. + (u_x^{n+1} - u_x^n)g_{u_x}(u^n, u_x^n, u_y^n) + (u_y^{n+1} - u_y^n)g_{u_y}(u^n, u_x^n, u_y^n) \right] d\xi d\eta.$$

Using (27), we get

$$\max_{x,y} |u_x^{n+1}| \leq mK_2b^2 \left[1 + \max_{x,y} |u^{n+1}| + \max_{x,y} |u^n| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_x^n| \right. \\ \left. + \max_{x,y} |u_y^{n+1}| + \max_{x,y} |u_y^n| \right], \quad (29)$$

and

$$\max_{x,y} |u_y^{n+1}| \leq mK_3b^2 \left[1 + \max_{x,y} |u^{n+1}| + \max_{x,y} |u^n| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_x^n| \right. \\ \left. + \max_{x,y} |u_y^{n+1}| + \max_{x,y} |u_y^n| \right]. \quad (30)$$

Adding (28), (29), and (30), we get

$$\max_{x,y} |u^{n+1}| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_y^{n+1}| \leq 1 + m(K_1 + K_2 + K_3)b^2 \left[1 + \max_{x,y} |u^{n+1}| + \max_{x,y} |u^n| \right. \\ \left. + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_x^n| + \max_{x,y} |u_y^{n+1}| + \max_{x,y} |u_y^n| \right].$$

Consequently, for $m(K_1 + K_2 + K_3)b^2 < 1$, we have

$$\max_{x,y} |u^{n+1}| + \max_{x,y} |u_x^{n+1}| + \max_{x,y} |u_y^{n+1}| \leq \frac{1}{1 - m(K_1 + K_2 + K_3)b^2} \left[1 + \right. \\ \left. m(K_1 + K_2 + K_3)b^2 \left(\max_{x,y} |u^n| + \max_{x,y} |u_x^n| + \max_{x,y} |u_y^n| \right) \right]. \quad (31)$$

Setting $n = 0$ in (31), letting $m_1 = \max_{x,y} |u^1| + \max_{x,y} |u_x^1| + \max_{x,y} |u_y^1|$, and using the fact that $|u^0(x, y)| \leq 2$, $|u_x^0(x, y)| \leq 1$, and $|u_y^0(x, y)| \leq 1$ for all $(x, y) \in [0, b]^2$, we get

$$m_1 \leq \frac{1}{1 - m(K_1 + K_2 + K_3)b^2} \left(1 + 4m(K_1 + K_2 + K_3)b^2 \right).$$

This latter upper bound is ≤ 2 , when $6m(K_1 + K_2 + K_3)b^2 \leq 1$.

As a result, we have $\max_{x,y} |u^1| \leq m_1 \leq 2$ under the condition that $0 \leq b \leq \frac{1}{\sqrt{6m(K_1 + K_2 + K_3)}}$, which can be easily satisfied by taking b small enough, i.e., $b \leq \frac{1}{\sqrt{6m(K_1 + K_2 + K_3)}}$.

The above procedure can be continued indicatively with the result which we can assert that $|u^n(x, y)| \leq 2$ for $(x, y) \in [0, b]^2$, under the condition that $b \leq \frac{1}{\sqrt{6m(K_1 + K_2 + K_3)}}$. As a result, we conclude that $\{u^n\}$ is well defined when b is sufficiently small. $\square$

Now, let's study the convergence of $\{u^n\}$. For that purpose, let's first define the following maximum norm

$$\|v\| = \max_{(x,y) \in [a,b]^2} |v(x, y)| + \max_{(x,y) \in [a,b]^2} |v_x(x, y)| + \max_{(x,y) \in [a,b]^2} |v_y(x, y)|,$$

which will be used to compare the following two errors: $\|u - u^{n+1}\|$ at iteration $n + 1$, and $\|u - u^n\|$ at the previous iteration n . Then, we establish the following theorem proving the quadratic convergence of $u^n(x, y)$ to the exact solution $u(x, y)$.

Theorem 4.2. *If the function $g(u, u_x, u_y) \in C_b^2(\mathbb{R})$, then the sequence $\{u^n\}$ defined by Equation (24) converges quadratically under the condition that $\hat{K}b^2m < 1$, i.e.,*

$$\|u - u^{n+1}\| \leq \frac{3\hat{K}Mb^2}{2(1 - \hat{K}mb^2)} \|u - u^n\|^2, \quad (32)$$

where m and M are two constants defined by

$$m = \max \left(\max_D |g|, \max_D |g_u|, \max_D |g_{u_x}|, \max_D |g_{u_y}| \right),$$

and

$$M = \max \left(\max_D |g_{uu}|, \max_D |g_{uu_x}|, \max_D |g_{uu_y}|, \max_D |g_{u_x u_y}| \right),$$

where $D = \{(u, u_x, u_y) \mid |u| \leq 2, |u_x| \leq 1, |u_y| \leq 1\}$. The constant $\hat{K} = K_1 + K_2 + K_3$, where K_1, K_2, K_3 are defined in (27).

Proof. First, we mention that the solution of the boundary value problem (23) can be expressed as

$$u(x, y) = 1 + \int_0^b \int_0^b K(x, y; \xi, \eta) g(u, u_x, u_y) d\xi d\eta. \quad (33)$$

Then, we subtract Equation (26) from Equation (33) and let $e^n = u - u^n$, which leads to

$$\begin{aligned} e^{n+1} &= \int_0^b \int_0^b K(x, y; \xi, \eta) [g(u) - g(u^n) - (u^{n+1} - u^n)g_u(u^n, u_x^n, u_y^n) \\ &\quad - (u_x^{n+1} - u_x^n)g_{u_x}(u^n, u_x^n, u_y^n) - (u_y^{n+1} - u_y^n)g_{u_y}(u^n, u_x^n, u_y^n)] d\xi d\eta \\ &= \int_0^b \int_0^b K(x, y; \xi, \eta) [g(u) - g(u^n) + (e^{n+1} - e^n)g_u(u^n, u_x^n, u_y^n) \\ &\quad + (e_x^{n+1} - e^n)g_{u_x}(u^n, u_x^n, u_y^n) + (e_y^{n+1} - e^n)g_{u_y}(u^n, u_x^n, u_y^n)] d\xi d\eta. \end{aligned}$$

Next, we apply Taylor's Theorem to obtain

$$\begin{aligned} e^{n+1} &= \int_0^b \int_0^b K(x, y; \xi, \eta) \left[\frac{(e^n)^2}{2} g_{uu}(v_n, w_n, z_n) + \frac{(e_x^n)^2}{2} g_{u_x u_x}(v_n, w_n, z_n) \right. \\ &\quad + \frac{(e_y^n)^2}{2} g_{u_y u_y}(v_n, w_n, z_n) + e^n e_x^n g_{u u_x}(v_n, w_n, z_n) + e^n e_y^n g_{u u_y}(v_n, w_n, z_n) \\ &\quad + e_x^n e_y^n g_{u_x u_y}(v_n, w_n, z_n) + e^{n+1} g_u(u^n, u_x^n, u_y^n) + e_x^{n+1} g_{u_x}(u^n, u_x^n, u_y^n) \\ &\quad \left. + e_y^{n+1} g_{u_y}(u^n, u_x^n, u_y^n) \right] d\xi d\eta, \end{aligned} \quad (34)$$

where v_n lies between u^n and u , w_n lies between u_x^n and u_x , and z_n lies between u_y^n and u_y .

Let

$$m = \max \left(\max_D |g|, \max_D |g_u|, \max_D |g_{u_x}|, \max_D |g_{u_y}| \right),$$

and

$$M = \max \left(\max_D |g_{uu}|, \max_D |g_{u u_x}|, \max_D |g_{u u_y}|, \max_D |g_{u_x u_y}| \right),$$

where $D = \{(u, u_x, u_y) \mid |u| \leq 2, |u_x| \leq 1, |u_y| \leq 1\}$. Then (34) implies

$$\begin{aligned} |e^{n+1}| &\leq K_1 \int_0^b \int_0^b \left[\frac{M}{2} (e^n)^2 + \frac{M}{2} (e_x^n)^2 + \frac{M}{2} (e_y^n)^2 + M |e^n| |e_x^n| + M |e^n| |e_y^n| \right. \\ &\quad \left. + M |e_x^n| |e_y^n| + m |e^{n+1}| + m |e_x^{n+1}| + m |e_y^{n+1}| \right] d\xi d\eta. \end{aligned}$$

Applying Young's inequality, we get

$$|e^{n+1}| \leq K_1 \int_0^b \int_0^b \left[\frac{3M}{2} (e^n)^2 + \frac{3M}{2} (e_x^n)^2 + \frac{3M}{2} (e_y^n)^2 \right. \\ \left. + m |e^{n+1}| + m |e_x^{n+1}| + m |e_y^{n+1}| \right] d\xi d\eta.$$

As a result, the following estimate is obtained

$$\max_{x,y} |e^{n+1}| \leq \frac{3K_1 Mb^2}{2} \left[\max_{x,y} |e^n|^2 + \max_{x,y} |e_x^n|^2 + \max_{x,y} |e_y^n|^2 \right] \\ + K_1 mb^2 \left[\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \right]. \quad (35)$$

Taking the derivatives of (34) with respect to x and y then repeating the same steps used to prove (35), we get

$$\max_{x,y} |e_x^{n+1}| \leq \frac{3K_2 Mb^2}{2} \left[\max_{x,y} |e^n|^2 + \max_{x,y} |e_x^n|^2 + \max_{x,y} |e_y^n|^2 \right] \\ + K_2 mb^2 \left[\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \right], \quad (36)$$

and

$$\max_{x,y} |e_y^{n+1}| \leq \frac{3K_3 Mb^2}{2} \left[\max_{x,y} |e^n|^2 + \max_{x,y} |e_x^n|^2 + \max_{x,y} |e_y^n|^2 \right] \\ + K_3 mb^2 \left[\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \right]. \quad (37)$$

Adding (35), (36), and (37), we arrive at

$$\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \leq \frac{3\hat{K} Mb^2}{2} \left[\max_{x,y} |e^n|^2 + \max_{x,y} |e_x^n|^2 + \max_{x,y} |e_y^n|^2 \right] \\ + \hat{K} mb^2 \left[\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \right], \quad (38)$$

where $\hat{K} = K_1 + K_2 + K_3$.

Then, for $\hat{K} b^2 m < 1$, we get

$$\max_{x,y} |e^{n+1}| + \max_{x,y} |e_x^{n+1}| + \max_{x,y} |e_y^{n+1}| \leq \frac{3\hat{K} Mb^2}{2(1 - \hat{K} mb^2)} \left[\max_{x,y} |e^n|^2 + \max_{x,y} |e_x^n|^2 + \max_{x,y} |e_y^n|^2 \right] \\ \leq \frac{3\hat{K} Mb^2}{2(1 - \hat{K} mb^2)} \left[\max_{x,y} |e^n| + \max_{x,y} |e_x^n| + \max_{x,y} |e_y^n| \right]^2,$$

or equivalently,

$$\|u - u^{n+1}\| \leq \frac{3\hat{K} Mb^2}{2(1 - \hat{K} mb^2)} \|u - u^n\|^2. \quad (39)$$

Hence, the sequence $\{u^n\}$ converges quadratically provided that $\hat{K} b^2 m < 1$, i.e., for $0 \leq b < \frac{1}{\sqrt{\hat{K} m}}$.

If we define $\beta = \frac{3\hat{K} Mb^2}{2(1 - \hat{K} mb^2)}$, then Equation (39) becomes

$$\|u - u^n\| \leq \beta \|u - u^{n-1}\|^2, \quad n = 1, 2, \dots$$

Using simple induction, we can easily deduce that

$$\|u - u^n\| \leq \beta^{2^{n-1}} \|u - u^0\|^{2^n} = \frac{1}{\beta} \left(\beta \|u - u^0\| \right)^{2^n}, \quad n = 1, 2, \dots,$$

which proves that the convergence depends on

$$\beta \|u - u^0\| = \frac{3\hat{K}Mb^2}{2(1 - \hat{K}mb^2)} \|u - u^0\|.$$

Hence, for an initial guess u^0 enough close to u , the sequence $\{u^n(x, y)\}$ converges uniformly to the function $u(x, y)$ which satisfies Equation (33). For instance, when $\|u - u^0\| < 1/\beta$.

Finally, from the estimate $\|u - u^n\| \leq \beta \|u - u^{n-1}\|^2$, we conclude the uniqueness of the sequence of approximations. $\square$

Next, we show that the sequence $\{V_{i,j}^n\}$ defined in (14) converges to the solution v_n of the linear equation (12). More precisely, we can show the following estimate.

Theorem 4.3. *Let v_n be the solution of (12) and $\{V_{i,j}^n\}$ be the finite difference solution given by (14). Then there exists a constant C independent of h such that*

$$|v^{n+1}(x_i, y_j) - V^{n+1}(x_i, y_j)| \leq Ch^2. \quad (40)$$

Proof. The proof of the result is elementary. Hence we omit it to save space. $\square$

Finally, using the triangle inequality, we deduce that the sequence $\{V_{i,j}^n\}$ converges to the solution $v(x_i, y_j)$ of the nonlinear equation (7).

Theorem 4.4. *Let v be the solution of (7), v^n be the solution of (12), and $\{V_{i,j}^n\}$ be the finite difference solution given by (14). Then $V^n(x_i, y_j)$ converges to the exact solution $v(x_i, y_j)$.*

Proof. By the triangle inequality, we have

$$|v(x_i, y_j) - V^n(x_i, y_j)| \leq |v(x_i, y_j) - v^n(x_i, y_j)| + |v^n(x_i, y_j) - V^n(x_i, y_j)| \leq Ch^2.$$

Since $v^n(x_i, y_j)$ converges to $v(x_i, y_j)$ and $V^n(x_i, y_j)$ converges to $v^n(x_i, y_j)$, we deduce that $V^n(x_i, y_j)$ converges to the exact solution $v(x_i, y_j)$. $\square$

We note that the approximation $V^n(x_i, y_j)$ of the solution $v(x_i, y_j)$ of the problem (7) consists of two steps:

1. Approximate $v(x_i, y_j)$ by $v^n(x_i, y_j)$, where $v^n(x_i, y_j)$ is the solution of (12).
2. Approximate $v^n(x_i, y_j)$ by the finite difference solution $V^n(x_i, y_j)$ defined in (14).

We proved that $v^n(x_i, y_j)$ converges to $v(x_i, y_j)$ quadratically and the solution $V^n(x_i, y_j)$ converges to $v^n(x_i, y_j)$ at $\mathcal{O}(h^2)$ rate. Therefore, $V^n(x_i, y_j)$ converges to $v(x_i, y_j)$.

5. Numerical simulation results

We display now various numerical simulations showing the two-branched computed solutions of the two-dimensional Bratu's problem for different cases. We first exhibit, in Fig. 2, both the upper and the lower branches of the numerical solution using various values of the transition parameter: $\lambda = 0.1, 1, 3$, and 6 . We also present in Fig. 3 the unique solution of the problem in the case of the computed critical value: $\lambda_c = 6.8083545$.

We further plot, in Fig. 4, the infinity norm of both branches of Bratu's problem solution for different values of λ . Such figure perfectly shows the bifurcated behavior of the solution. It numerically confirms the known theoretical facts about Bratu's problem.

Additionally, in Table 1, we present the maximum value of the lower branch (i.e. $\|U_\lambda^{Lower}\|_\infty$) and the maximum value of upper branch (i.e. $\|U_\lambda^{Upper}\|_\infty$) of the computed solution of the two-dimensional Bratu's problem. Taking a close look at the aforementioned results, we can conclude that the infinity norm of the lower branch of the numerical solution $\|U_\lambda^{Lower}\|_\infty$ increases, when λ increases. Its maximum is attained when $\lambda = \lambda_c$. Contrarily, the infinity norm of the upper branch $\|U_\lambda^{Upper}\|_\infty$ decreases, when λ increases. Its minimum is attained when $\lambda = \lambda_c$.

We also exhibit in the same Table 1 a comparison between our computed results and one of the most recent and accurate works available in the literature, by Liu et al. [27]. It is clear that there is an agreement between the results computed by the iterative finite difference method proposed in this manuscript and the results computed by the Coiflet-type wavelet expansion method [27], for both the lower and upper branches of the solution ($\|U_\lambda^{Lower}\|_\infty$ and $\|U_\lambda^{Upper}\|_\infty$).

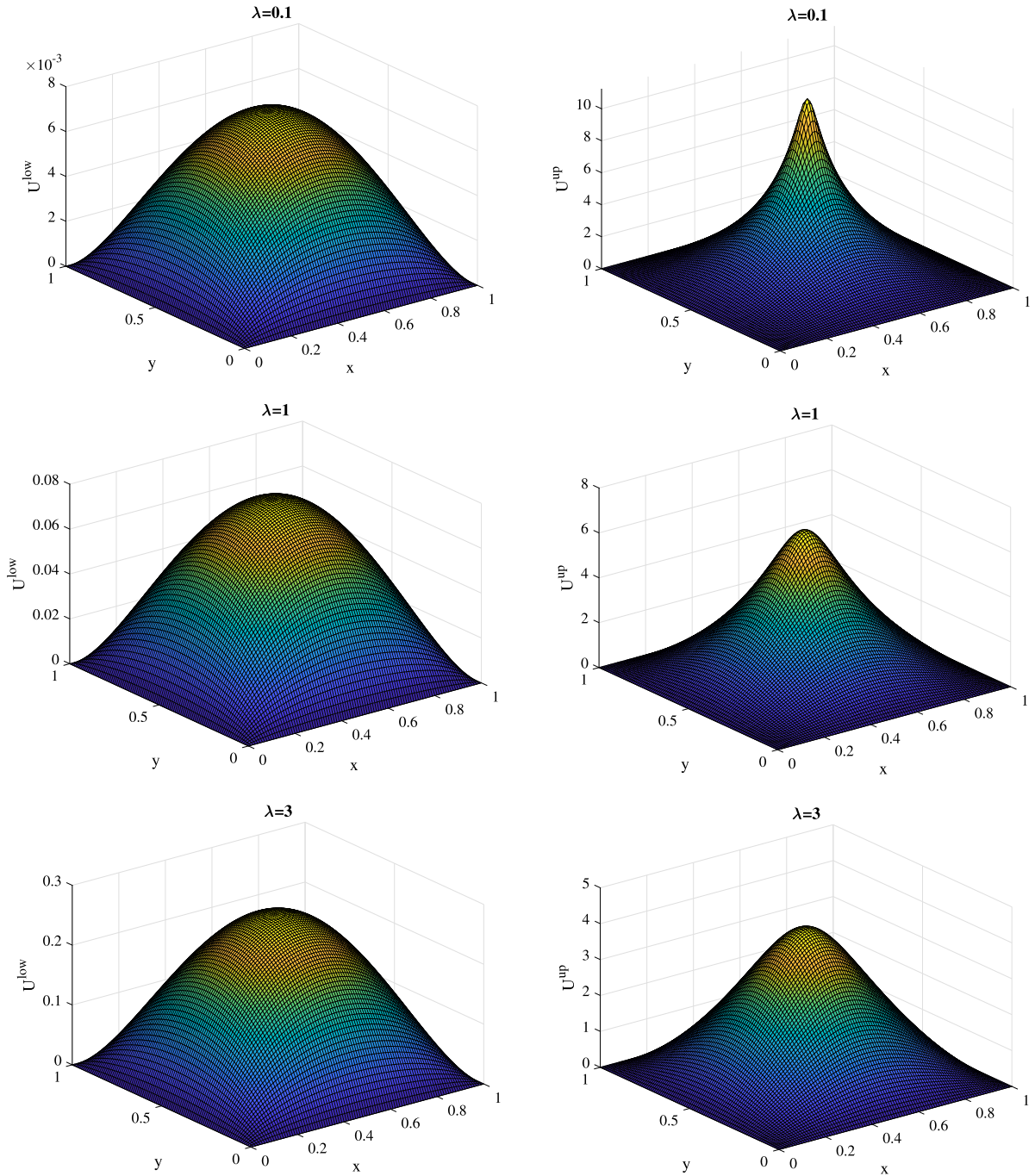


Fig. 2. The lower computed solution U^{low} and the upper computed solution U^{up} of Bratu's problem.

We also present in Table 2 the critical value λ_c of the transition parameter computed by our proposed IFD method, compared to different estimations of λ_c computed by some of the available works about the two-dimensional Bratu's problem. We also exhibit in the same Table 2 the infinity norm $\|U_{\lambda_c}\|_{\infty}$ of the unique solution when $\lambda = \lambda_c$. It is remarkable that the critical value of the transition parameter $\lambda_c = 6.8083545$ computed by the method presented in this manuscript is in perfect agreement with many references available in the literature such as [16,18,19,30] (for instance, $\lambda_c = 6.808124423$ in [18,30]). However, as mentioned in the introduction section of this manuscript, some other numerical methods have lead to slightly different values of the critical value of the transition parameter, such as the work of Odejide et al. [31] in which the authors predicted that: $6.939571823 \leq \lambda_c \leq 7.12222$. Indeed, finding the **exact** critical value of λ_c is still an open question for this problem.

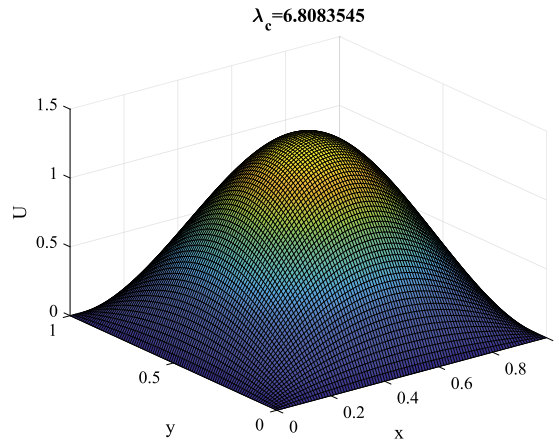


Fig. 3. The critical computed solution of Bratu's problem.

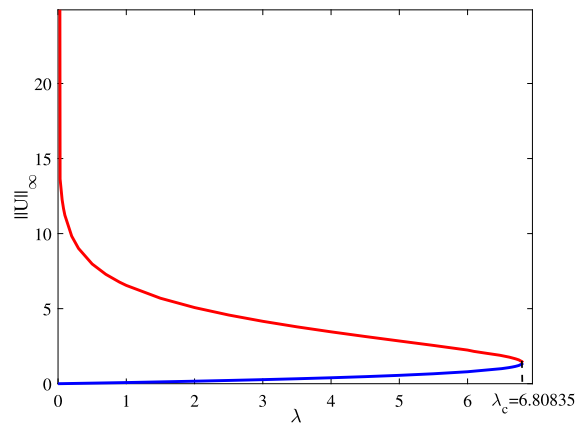


Fig. 4. Bifurcated nature of the computed solution to the two-dimensional Bratu's problem.

Table 1

The maximum of the lower and the upper branches of the computed solution for the two-dimensional Bratu's problem compared to results in [27].

λ	$\ U_{\lambda}^{Lower}\ _{\infty}$	$\ U_{\lambda}^{Lower}\ _{\infty}$ [27]	$\ U_{\lambda}^{Upper}\ _{\infty}$	$\ U_{\lambda}^{Upper}\ _{\infty}$ [27]
0.1	$7.407507701611240 \times 10^{-3}$	$7.408093229 \times 10^{-3}$	11.22544470616899	11.38218777
1.0	$7.809460633062898 \times 10^{-2}$	$7.810101876 \times 10^{-2}$	6.548562494506136	6.548494384
2.0	$1.668813508714819 \times 10^{-1}$	$1.668957564 \times 10^{-1}$	5.072228156071850	5.072459550
3.0	$2.703415213530049 \times 10^{-1}$	$2.703663546 \times 10^{-1}$	4.156576781032640	4.156756192
4.0	$3.954865464642406 \times 10^{-1}$	$3.955259988 \times 10^{-1}$	3.454818156389594	3.454936050
5.0	$5.568969391509362 \times 10^{-1}$	$5.569597638 \times 10^{-1}$	2.845889753560470	2.845945007
6.0	$7.969945370409719 \times 10^{-1}$	$7.971089810 \times 10^{-1}$	2.239940006850879	2.239910842
6.8	1.322254309273867	1.323450119	1.463403698314159	1.462270493

6. Conclusion

In this manuscript, we presented and analyzed a new iterative finite difference scheme to approximate both branches of a two-branched solution to the famous two-dimensional Bratu's problem. We first defined a new transformation of Bratu's problem conserving the solution bifurcated behavior. Then, using Newton-Raphson-Kantorovich approximation in function space [7], we extended the iterative finite difference method developed in [8] to the two-dimensional case, in order to solve the two-dimensional Bratu's problem. We have also performed a convergence analysis proving that our algorithm converges quadratically to the exact solution of the problem. Finally, our numerical simulation results have confirmed the capability of our algorithm in accurately computing the two-branches of the numerical solution of the two-dimensional Bratu's problem, for various cases of the transition parameter λ .

Table 2

The estimated values of the transition parameter λ_c for the two-dimensional Bratu's problem computed by various available methods, and the infinity norm of the unique solution computed by each method (if available) for $\lambda = \lambda_c$.

Method	Computed λ_c	$\ U_{\lambda_c}\ _{\infty}$
Iterative finite difference method (Current work)	$\lambda_c = 6.8083545$	1.396844181
Arc-length continuation & multi-grid techniques [15]	6.80811698	1.3916603
Branch through a turning point with a Newton like method [2]	6.80811698	1.3916603
Collocation Methods [18]	6.808124423	N/A
Radial Operators Method [22]	6.808124423	N/A
Multigrid-Lanczos algorithm [16]	6.808124423	N/A
Multigrid-based method with new smoother [30]	6.808124423	N/A
Nonstandard finite-difference (NSFD) [29]	6.808124423	1.3566
Modified inverse interpolation [33]	6.80817	1.2732
Coiflet-type wavelet expansion [27]	$6.80 < \lambda < 6.81$	N/A
Multiquadric Discretization Method [19]	6.827400	N/A
Weighted Residual Method [31]	6.7796325	1.369513771
Near exact solution [31]	7.027661438	1.1868
Finite difference method [31]	7.12222	1.3565527
Iterative differential quadrature [32]	7.027661438	1.186842168
Hybrid orthonormal Bernstein & block-pulse wavelet [5]	7.027661438	N/A

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References

- [1] S. Abbasbandy, M. Hashemi, C. Liu, The Lie-group shooting method for solving the Bratu equation, *Commun. Nonlinear Sci. Numer. Simul.* 16 (2011) 4238–4249.
- [2] J.P. Abbott, An efficient algorithm for the determination of certain bifurcation points, *J. Comput. Appl. Math.* 4 (1978) 19–27.
- [3] M.B. AbdelMalek, A. Abdelrazek, M. Ghazy, G. Gamal, A modified perturbation solution to the one-dimensional Bratu problem, *Appl. Math. Comput.* 354 (2019) 296–304.
- [4] Y. Aksoy, M. Pakdemirli, New perturbation–iteration solutions for Bratu-type equations, *Comput. Math. Appl.* 59 (8) (2010) 2802–2808.
- [5] M.R. Ali, A.R. Hadhoud, Hybrid orthonormal Bernstein and block-pulse functions wavelet scheme for solving the 2D Bratu problem, *Results Phys.* 12 (2019) 525–530.
- [6] R.E. Bank, T.F. Chart, PLTMGC: a multi-grid continuation program for parameterized nonlinear elliptic systems, *SIAM J. Sci. Stat. Comput.* 7 (1986) 540–559.
- [7] R.E. Bellman, R.E. Kalaba, *Quasilinearization and Nonlinear Boundary-Value Problems*, American Elsevier Publishing Company, New York, 1965.
- [8] M. Ben-Romdhane, H. Temimi, M. Baccouch, An iterative finite difference method for approximating the two-branched solution of Bratu's problem, *Appl. Numer. Math.* 139 (2019) 62–76.
- [9] J. Boyd, One-point pseudospectral collocation for the one-dimensional Bratu equation, *Appl. Math. Comput.* 217 (2011) 5553–5565.
- [10] J.P. Boyd, An analytical and numerical study of the two-dimensional Bratu equation, *J. Sci. Comput.* 1 (1986) 183–206.
- [11] G. Bratu, Sur les equation integrals non-lineaires, *Bull. Soc. Math. Fr.* 42 (1914) 113–142.
- [12] R. Buckmire, Application of a Mickens finite-difference scheme to the cylindrical Bratu–Gelfand problem, *Numer. Methods Partial Differ. Equ.* 20 (3) (2004) 327–337.
- [13] R. Buckmire, Applications of Mickens finite differences to several related boundary value problems, in: *Advances in the Applications of Nonstandard Finite Difference Schemes*, 2005, pp. 47–87.
- [14] H. Caglar, N. Caglar, M. Ozer, A. Valaristos, A.N. Anagnostopoulos, B-spline method for solving Bratu's problem, *Int. J. Comput. Math.* 87 (8) (2010) 1885–1891.
- [15] T.F. Chan, H.B. Keller, Arc-length continuation and multigrid techniques for nonlinear elliptic eigenvalue problems, *SIAM J. Sci. Stat. Comput.* 3 (1982) 173–194.
- [16] S.L. Chang, C.S. Chien, A multigrid-Lanczos algorithm for the numerical solutions of nonlinear eigenvalue problems, *Int. J. Bifurc. Chaos* 13 (2003) 1217–1228.
- [17] E. Deeba, S. Khuri, S. Xie, An algorithm for solving boundary value problems, *J. Comput. Phys.* 159 (2000) 125–138.
- [18] E.J. Doedel, H. Sharifi, Collocation Methods for Continuation Problems in Nonlinear Elliptic PDEs, *Notes on Numerical Fluid Mechanics*, vol. 74, 2000, pp. 105–118.
- [19] A.I. Fedoseyev, M.J. Friedman, E.J. Kansa, Continuation for nonlinear elliptic partial differential equations discretized by the multiquadratic method, *Int. J. Bifurc. Chaos* 10 (2000) 481–492.
- [20] I.M. Gelfand, Some problems in the theory of quasi-linear equations, *Amer. Math. Soc. Transl. Ser. 2* 29 (1963) 295–381. *Twelve Papers on Logic and Differential Equations*.
- [21] J. He, Some asymptotic methods for strongly nonlinear equations, *Int. J. Mod. Phys. B* 20 (2006) 1141–1199.
- [22] J. Jacobsen, K. Schmitt, The Liouville–Bratu–Gelfand problem for radial operators, *J. Differ. Equ.* 184 (2002) 283–298.
- [23] R. Jalilian, Non-polynomial spline method for solving Bratu's problem, *Comput. Phys. Commun.* 181 (2010) 1868–1872.
- [24] E. Keshavarz, Y. Ordokhani, M. Razzaghi, The Taylor wavelets method for solving the initial and boundary value problems of Bratu-type equations, *Appl. Numer. Math.* 128 (2018) 205–216.
- [25] S. Khuri, A new approach to Bratu's problem, *Appl. Math. Comput.* 147 (1) (2004) 131–136.
- [26] J. Liouville, Sur l'équation aux différences partielles $\frac{d^2 \log \lambda}{dudv} \pm \frac{\lambda}{2a^2} = 0$, *J. Math. Pures Appl.* 18 (1853) 71–72.
- [27] X. Liu, J. Wang, Y. Zhou, Wavelet solution of a class of two-dimensional nonlinear boundary value problems, *Comput. Model. Eng. Sci.* 92 (5) (2013) 493–505.
- [28] A. Mohsen, On the integral solution of the one-dimensional Bratu problem, *J. Comput. Appl. Math.* 251 (2013) 61–66.
- [29] A. Mohsen, A simple solution of the Bratu problem, *Comput. Math. Appl.* 67 (1) (2014) 26–33.

- [30] A. Mohsen, L.F. Sedeek, S.A. Mohamed, New smoother to enhance multigrid-based methods for Bratu problem, *Appl. Math. Comput.* 204 (2008) 325–339.
- [31] S.A. Odejide, Y.A. Aregbesola, A note on two dimensional Bratu problem, *Kragujev. J. Math.* 29 (2006) 49–56.
- [32] O. Ragab, L.F. Seddek, M.S. Matbully, Iterative differential quadrature solutions for Bratu problem, *Comput. Math. Appl.* 74 (2017) 249–257.
- [33] R.B. Simpson, A method for the numerical determination of bifurcation states of nonlinear systems of equations, *SIAM J. Numer. Anal.* 12 (3) (1975) 439–451.
- [34] M. Syam, A. Hamdan, An efficient method for solving Bratu equation, *Appl. Math. Comput.* 176 (2006) 704–713.
- [35] H. Temimi, M. Ben-Romdhane, An iterative finite difference method for solving Bratu's problem, *J. Comput. Appl. Math.* 292 (2016) 76–82.
- [36] S. Tomar, R.K. Pandey, An efficient iterative method for solving Bratu-type equations, *J. Comput. Appl. Math.* 357 (2019) 71–84.
- [37] Y. Wan, Q. Guo, N. Pan, Thermo-electro-hydrodynamic model for electrospinning process, *Int. J. Nonlinear Sci. Numer. Simul.* 5 (2004) 5–8.
- [38] Z. Yang, S. Liao, A HAM-based wavelet approach for nonlinear partial differential equations: two dimensional Bratu problem as an application, *Commun. Nonlinear Sci. Numer. Simul.* 53 (2017) 249–262.